CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Project Report | Python • PostgreSQL • Power BI

Understanding purchase patterns, customer segments, product preferences, and subscription behavior.

1. Project Overview

This project looks at customer shopping behavior using transaction data from 3,900 purchases across different product categories. The aim is to understand spending patterns, product preferences, customer groups, and subscription behavior, then use those findings to support business decisions.

3,900	18	37
Purchase records	Dataset columns	Missing review ratings

2. Dataset Summary

The dataset contains customer, purchase, and shopping-behavior details, including:

- **Customer details:** age, gender, location, and subscription status.
- **Purchase details:** item purchased, category, purchase amount, season, size, and color.
- **Shopping behavior:** discount use, previous purchases, purchase frequency, review rating, and shipping type.

Data Preparation

- Loaded the data into Python using pandas and checked its structure and summary statistics.
- Handled the 37 missing review ratings by filling them with the median rating for each product category.
- Renamed columns to snake_case to make them easier to use in code and SQL.
- Created an *age_group* field and a *purchase_frequency_days* feature.
- Checked discount and promo-code fields for redundancy and removed *promo_code_used*.
- Connected Python to PostgreSQL and loaded the cleaned data for SQL analysis.

3. SQL Analysis

PostgreSQL was used to answer business questions about customer spending, product performance, shipping, discounts, subscriptions, and customer loyalty.

Analysis	Question explored
Revenue by gender	How does total revenue compare between male and female customers?
High-spending discount users	Which customers used discounts but still spent above the average purchase amount?
Top five products by rating	Which products have the highest average review ratings?
Shipping type comparison	How do average purchase amounts compare for Standard and Express shipping?

Analysis	Question explored
Subscribers vs. non-subscribers	How do average spending and total revenue differ by subscription status?
Discount-dependent products	Which five products have the highest percentage of discounted purchases?
Customer segmentation	How can customers be grouped as New, Returning, or Loyal using purchase history?
Top products per category	What are the three most purchased products in each category?
Repeat buyers and subscriptions	Are customers with more than five purchases more likely to subscribe?
Revenue by age group	How much total revenue does each age group contribute?

4. Power BI Dashboard

An interactive Power BI dashboard was created to present the analysis visually. It gives users a way to explore the customer and purchase data through charts and report filters. The supplied reference confirms that a dashboard was built, but does not provide readable numeric chart results in the extracted text, so this report does not invent specific totals or rankings.

5. Key Business Areas

- **Subscriptions:** Compare subscribers with non-subscribers and review the relationship between subscription status and spending.
- **Customer loyalty:** Use purchase history to distinguish New, Returning, and Loyal customers.
- **Product performance:** Compare popular products, category-level purchases, and customer ratings.
- **Discounts:** Review products frequently purchased with discounts and examine high-spending discount users.
- **Customer segments:** Compare revenue by gender and age group.
- **Shipping:** Compare average purchase amounts for Standard and Express shipping.

6. Business Recommendations

- **Encourage subscriptions:** Promote useful subscriber benefits to encourage sign-ups.
- **Reward repeat customers:** Consider loyalty rewards for returning buyers and ways to strengthen retention.
- **Review discount strategy:** Check whether discounts support sales without unnecessarily reducing margins.
- **Promote strong products:** Feature highly rated and frequently purchased products in campaigns.
- **Use targeted marketing:** Focus campaigns on age groups that contribute strongly to revenue and consider the needs of Express-shipping customers.

7. Conclusion

This project combines Python data cleaning, PostgreSQL queries, and Power BI visualization to examine customer shopping behavior. The analysis is designed to help a business better understand customer segments, product preferences, discount use, shipping choices, and subscription behavior. The findings can guide further investigation and planning. Specific business actions should be checked against the actual query results and dashboard figures before implementation.

Tools Used

Python: data loading, cleaning, and feature engineering
PostgreSQL: business-focused SQL analysis
Power BI: interactive dashboard and visual reporting

Prepared from the supplied reference document, “Customer Shopping Behavior Analysis.”